

Tutorial 9 - Evaluation of RAG

Week 9 of INFS4205/7205 - Sem 1, 2026

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Recap from Lecture 9: RAGAS Retrieval Metrics

Retrieval metrics are still the first layer of evaluation in M-RAG

- **Non-Rank Based Metrics:** often assess binary outcomes

Accuracy?

$$\text{accuracy} = (tp + tn) / (tp + fp + fn + tn)$$

$$\text{Precision} = \frac{\#(\text{relevant items retrieved})}{\#(\text{retrieved items})} = P(\text{relevant}|\text{retrieved})$$

	Relevant	Nonrelevant
Retrieved	true positives (tp)	false positives (fp)
Not retrieved	false negatives (fn)	true negatives (tn)

Precision, Recall

fraction of retrieved documents that are relevant

$$P = tp / (tp + fp)$$

Recap from Lecture 9: RAGAS Retrieval Metrics

Retrieval metrics are still the first layer of evaluation in M-RAG

- ***Non-Rank Based Metrics***: often assess binary outcomes

$$\text{Recall} = \frac{\#(\text{relevant items retrieved})}{\#(\text{relevant items})} = P(\text{retrieved}|\text{relevant})$$

fraction of relevant documents that are retrieved

$$R = tp / (tp + fn)$$

Those are all for binary outcomes

Recap from Lecture 9: RAGAS Retrieval Metrics

Normalised discounted cumulative gain (nDCG@K)

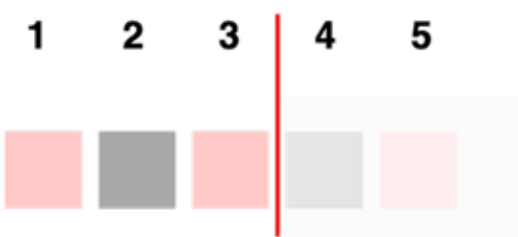
designed for situations of **non-binary** notions of relevance

$$\text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}$$

Discounted Cumulative Gain (DCG)

$$\text{DCG} = \sum_{i=1}^n \frac{rel_i}{\log_2(i+1)}$$

K = 3



$$\text{DCG@3} = \frac{1}{\log_2(1+1)} + \frac{0}{\log_2(1+2)} + \frac{1}{\log_2(1+3)}$$

$$\text{DCG@3} = 1 + 0 + 0.5 = 1.5$$

K = 3



$$\text{DCG@3} = \frac{1}{\log_2(1+1)} + \frac{1}{\log_2(1+2)} + \frac{0}{\log_2(1+3)}$$

$$\text{DCG@3} = 1 + \frac{1}{1.585} + 0 \approx 1.63$$

Recap from Lecture 9: RAGAS Retrieval Metrics

Normalised discounted cumulative gain (nDCG@K)

designed for situations of **non-binary** notions of relevance

$$\text{NDCG@K} = \frac{\text{DCG@K}}{\text{IDCG@K}}$$

Ideal Discounted Cumulative Gain (IDCG)

$$\text{IDCG@K} = \sum_{i=1}^K \frac{rel_i^*}{\log_2(i+1)}$$



$$\text{DCG@3} = \frac{2}{\log_2(1+1)} + \frac{0}{\log_2(2+1)} + \frac{1}{\log_2(3+1)} = 2.5$$

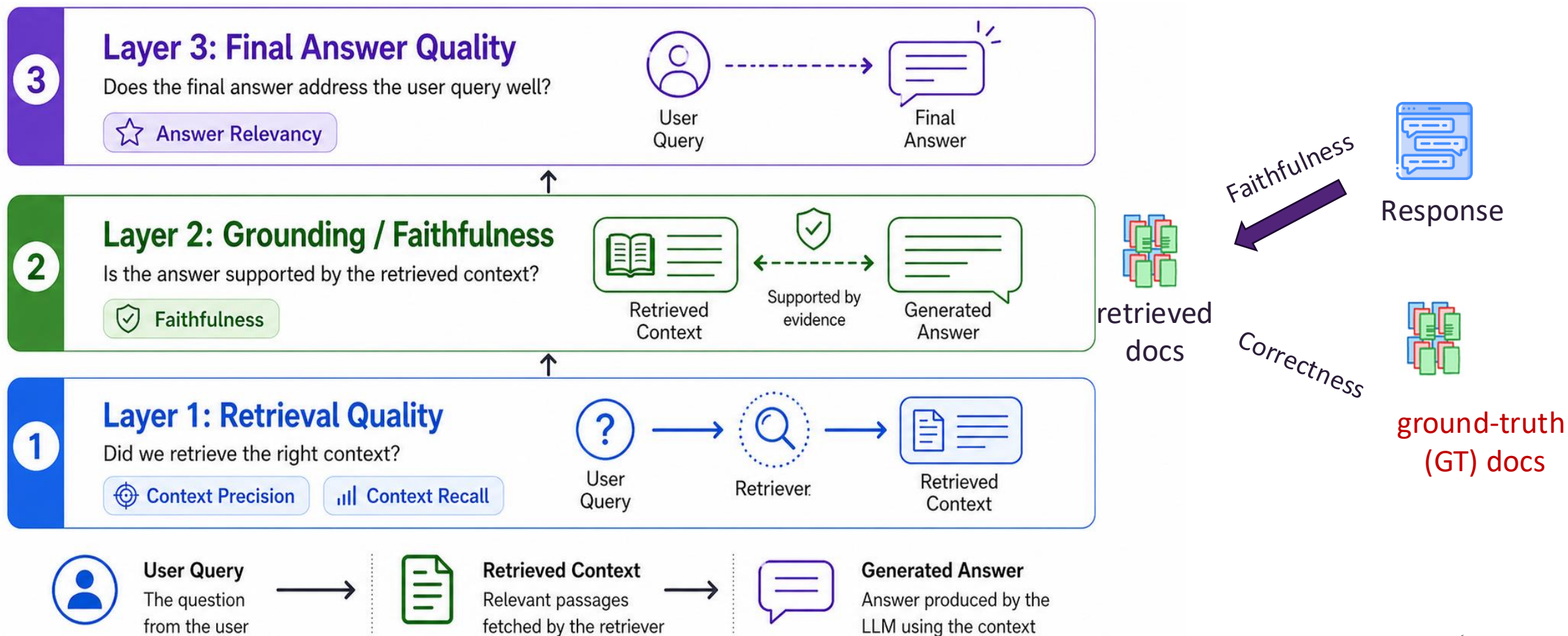


$$\text{IDCG@3} = \frac{2}{\log_2(1+1)} + \frac{1}{\log_2(2+1)} + \frac{0}{\log_2(3+1)} \approx 2.63$$

$$\text{NDCG@3} = \frac{\text{DCG@3}}{\text{IDCG@3}} = \frac{2.5}{2.63} \approx 0.95$$

Recap from Lecture 9: RAGAS Retrieval Metrics

What Retrieval Eval misses...



Retrieval-Augmented Generation Assessment

Q(a). What is a metric for retrieval quality in a RAG system?

Retrieval-Augmented Generation Assessment

Q(a). What is a metric for retrieval quality in a RAG system?

A. Recall@k or Context Recall.

It measures whether relevant evidence is retrieved in the top-k results, i.e., whether the system successfully retrieves the supporting information needed for the task.

In a word, we care about how well the generated answer aligns with the intent and content of the initial query.

Retrieval-Augmented Generation Assessment

Q(b). What is a metric for whether the answer is supported by retrieved evidence?

Retrieval-Augmented Generation Assessment

Q(b). What is a metric for whether the answer is supported by retrieved evidence?

A. Faithfulness or Groundedness.

It measures whether the generated answer is supported by the retrieved evidence rather than being hallucinated or invented by the model.

In a word, we care about whether we get coherent and contextually appropriate outputs from the retrieved documents.

Retrieval-Augmented Generation Assessment

Q(c). What is a metric for whether the answer addresses the user's question?

Retrieval-Augmented Generation Assessment

Q(c). What is a metric for whether the answer addresses the user's question?

A. Answer Accuracy or Answer Correctness.

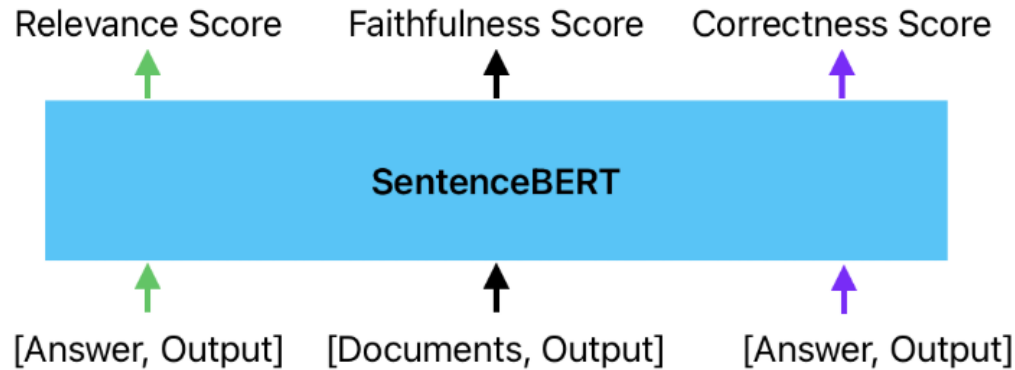
It measures whether the final answer matches the expected or ground-truth answer and whether it properly responds to the user's question rather than being off-topic or incomplete.

In a word, we care about whether it is a right answer.

Generation Evaluation

When we have ground-truth documents and answers for each test query

Use sentence Transformer to measure the **semantic similarity for each pair**



Use LLM as a judge for each pair

Do they semantically align with each other?

sentence 1: {golden answer}
sentence 2: {output}

Do they semantically align with each other?

Example #1:

sentence 1: This class is awesome.
sentence 2: I like this class.
answer: aligned

Example #2:

.....

Example #3:

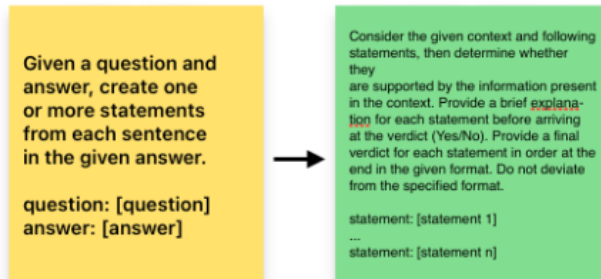
.....

sentence 1: {golden answer}
sentence 2: {output}
answer:

Generation Evaluation

RAGAs (+ FactScore[MKL23]) pipeline[EJL24] with LLM backends and the key idea: reverse engineering and decomposition

Faithfulness

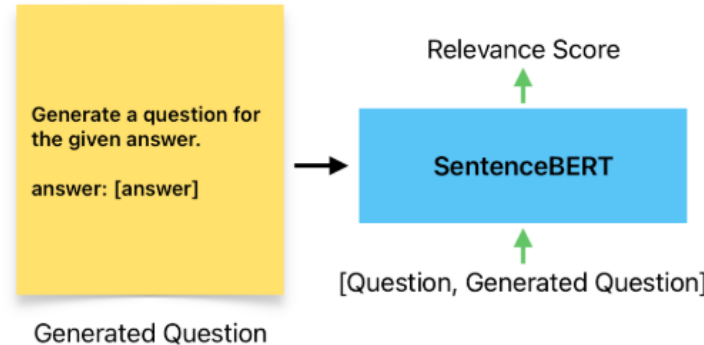


V statements

S verified statements

$$F = \frac{S}{V}$$

Answer Relevance



Generated Question

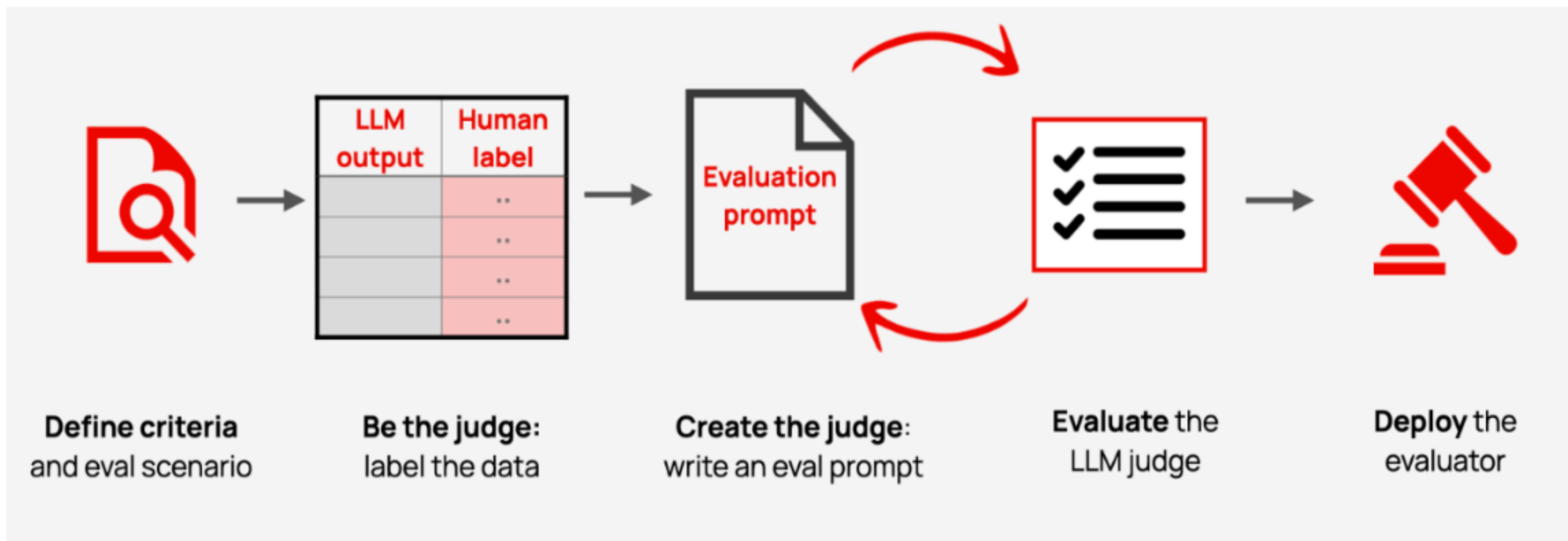
Context Relevance

Please extract relevant sentences from the provided context that can potentially help answer the following question. If no relevant sentences are found, or if you believe the question cannot be answered from the given context, return the phrase "Insufficient Information". While extracting candidate sentences you're not allowed to make any changes to sentences from given context.

$$CR = \frac{\text{number of extracted sentences}}{\text{total number of sentences in } c(q)}$$

where $c(q)$ is all retrieved documents

Practice for semi-Automated RAG Evaluation



RAGAs vs TruLens vs DeepEval

Dimension	RAGAS	TruLens	DeepEval
Focus	RAG evaluation	Eval + Tracing	Comprehensive LLM eval
Core Metrics	Faithfulness / Groundedness, Answer Accuracy / Answer Correctness, Recall@k / Context Recall, Context Precision	RAG Triad: Context relevance, Groundedness, Answer relevance etc	50+ Metrics: RAG metrics, agent metrics, safety metrics, multimodal metrics
LLM-as-Judge	Yes	Yes	Yes
G-Eval Support	No native support	Via feedback prompt	Native G-Eval metric
Tracing	Minimal	OpenTelemetry tracing	Function-level tracing
Custom Metrics	Limited	Feedback API	DAG, G-Eval, Custom
Best For	Fast RAG iteration	Agent Debugging	Production QA
Shared Characteristics			
Layer	Inference-layer evaluation only		
Reference-Free	LLM-as-a-Judge supported		
Limitation	Cannot fully assess retrieval index trustworthiness (only assess RAG output)		

Next Week

- Era of Agents

